

CHANG KAI WEN

SOFTWARE ENGINEER • DATA & MACHINE LEARNING

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PROFILE

Early-career software engineer and data scientist with one year of professional experience at GovTech, complemented by internships across consulting, utilities and applied AI. Experienced in LLM evaluation, guardrails, machine-learning models, data pipelines and full-stack development. Brings a practical foundation in computing science and a track record of building technology for public-sector and operational use cases.

EXPERIENCE

Software Engineer / Data Scientist | Government Technology Agency (GovTech) **Mar 2025 – Feb 2026**

- Developed a chatbot proof of concept by adapting an open-source web application and large language models to the operational needs of government officers.
- Delivered an LLM evaluation API suite supporting evaluation job creation, execution-status monitoring and cancellation, streamlining evaluation workflows.
- Enhanced an off-topic guardrail to improve robustness and generalisation across a wider range of user prompts.

Software Engineer Intern | Accenture **Aug 2023 – Apr 2024**

- Contributed to a large-scale cloud system-integration project in the health and public-services sector.
- Implemented clearer frontend error messages and standardised component naming conventions to improve usability and code readability.
- Prepared and managed test data for feature validation, supporting production-readiness checks for web application releases.

Data Engineer / Scientist Intern | SP Digital, SP Group **Jul 2018 – Feb 2019**

- Developed a machine-learning model using utility-consumption data to detect residential move-in events and support more efficient metering-to-billing operations.
- Built Talend and Apache Hive data pipelines for automated test reporting and migration between Hadoop clusters.

Software Developer | Capability Development, Temasek Polytechnic **May 2019 – Jul 2019**

- Enhanced a self-registration system with face-recognition capabilities for a public-facing technology showcase.
- Developed a mobile accessibility application that converted menu images to text and read the content aloud for visually impaired users.

Administrative Support Assistant | HQCDO, Singapore Armed Forces **Oct 2019 – Aug 2021**

- Managed IT assets and controlled secure server-room access in support of commando operations.

EDUCATION

BSc (Hons) Computing Science | Singapore Institute of Technology & University of Glasgow | 2021–2024

Diploma in Information Technology | Temasek Polytechnic | 2016–2019

SELECTED PROJECTS

AI-driven avatar animation platform: Built a platform that creates talking digital avatars from source videos and images.

Machine learning in C: Implemented a classification algorithm for semen-analysis data to support fertility diagnostics and medical decision-making.

Language-learning Android app: Created a Mandarin-learning application with translation, pronunciation and Hanyu Pinyin features for English speakers.

Full-stack publishing platform: Built a blog-creation application using the MEAN stack for individual and organisational content sharing.

Business intelligence and big data: Designed Tableau dashboards and performed ETL workflows with Talend to support operational decision-making.

TECHNICAL SKILLS

Languages	Python, JavaScript/TypeScript, Java, C#, C, HTML, CSS, Assembly (Raspberry Pi 4)
Data & ML	Pandas, NumPy, scikit-learn, XGBoost, TensorFlow, Jupyter Notebook, Google Colab
Web & application	Node.js, Express.js, Angular, Ionic, Flask, ASP.NET MVC, ASP.NET Web API, Android Studio
Data platforms	MongoDB, MySQL, Talend for Big Data, Apache Hive, Hadoop, Pig, Tableau, QlikView